# FTC Shooter Physics

Guided Problem Set · Parts A–E

Name: _____ Date: _____

**How to use this set**

By the end you will be able to explain how fast a ball leaves a single-flywheel FTC shooter, and why the hood's grip matters. Each part adds one tool you need.

- **Gravity:** use $g = 9.8\ m/s^2$.
- **Always draw a free-body diagram** for any problem that asks for a force or an acceleration.
- **Sign convention:** “forward” is the direction the ball travels out of the shooter. Pick a direction for any unknown force; a negative answer means it points the other way.
- **Shared numbers:** several problems reuse the same ball and shooter, so you can watch one model get built up piece by piece.

**P0. Make a prediction before you start.** Our shooter hoods are 3D printed and fairly slick. Would a *lower*-friction hood make the ball leave faster, slower, or the same? Write your prediction and one sentence of reasoning. You will check it in E6.

# Part A: Friction

Friction is the only force that pushes the ball forward in a shooter, so it is where we start. Static friction adjusts to whatever is needed, up to a maximum; kinetic friction has a fixed size once surfaces slide.

$$f_s \le \mu_s N \qquad f_k = \mu_k N$$

**Read more**
The Physics Classroom: Types of Forces (friction section) · College Physics 5.1: Friction

**A1. Push a block.** A 2.0 kg block rests on a level table with $\mu_s = 0.50$ and $\mu_k = 0.40$.

**A1.1** You push horizontally with 5.0 N. Does the block move? What is the friction force?

**A1.2** You push with 12 N instead. Does it move? Find the friction force and the acceleration.

**A2. Box on a conveyor.** A box is set down at rest on a belt moving at 3.0 m/s. $\mu_k = 0.30$ between box and belt.

**A2.1** Which way does friction act on the box? Explain using the *relative* motion of the surfaces.

**A2.2** Find the box's acceleration while it slips.

**A2.3** How long until the box stops slipping?

**A2.4** How far does the box travel before it stops slipping?

**A2.5** What is the friction force on the box after it stops slipping?

**A2.6** In our model, kinetic friction does not depend on how fast the surfaces slide. What *would* change if the belt ran at 6 m/s instead of 3 m/s? What would stay the same?

**A3. Pinched block.** A 0.10 kg block is held off the ground only by being squeezed between two vertical plates. Each plate presses with normal force N, and $\mu_s = 0.60$ at both plates.

**A3.1** Draw the free-body diagram.

**A3.2** Find the smallest N that keeps the block from falling.

**A3.3** What happens to the friction force if you squeeze twice as hard as that minimum?

## Part B: Springs and normal force

A squeezed foam ball pushes back like a spring, and that push is the normal force that lets friction grip it. Compression δ is how much smaller the gap is than the ball.

$$F = k\delta \qquad U_{spring} = \tfrac{1}{2} k \delta^2$$

**Read more**

The Physics Classroom: Potential Energy (springs) · The Physics Classroom: Kinetic Energy
College Physics 16.1: Hooke's Law · College Physics 7.2: Kinetic Energy

*This problem set uses δ for the compression instead of x. College Physics writes F = −kx; the minus sign only says the spring pushes back against the squeeze.*

**B1. One-point test.** A 1.0 kg mass resting on a foam ball compresses it by 6.0 mm. Find the ball's spring constant k in N/m.

**B2. Squeeze test data.** A team stacks masses on a ball and measures the compression.

| Mass on ball (kg) | Compression δ (mm) | Force (N) | Compression (m) |
|---|---|---|---|
| 0.5 | 3.0 | | |
| 1.0 | 6.1 | | |
| 1.5 | 8.9 | | |
| 2.0 | 12.2 | | |

**B2.1** Convert each row to force (N) and compression (m). Fill in the last two columns.

**B2.2** Graph force vs. compression (Desmos is fine). Is the ball's response close to linear?

**B2.3** Find k from the slope of a best-fit line through the origin.

**B3. From gap to normal force.** The ball's diameter is 127 mm. Use k = 1600 N/m.

**B3.1** The gap between the wheel surface and the hood is 115 mm. Find δ and the normal force N.

**B3.2** A student adds a shim so the gap becomes 119 mm. Find the new δ and N.

**B3.3** Write a general formula for N in terms of the ball diameter D, the gap G, and k.

**B4. Is the ball a spring launcher?** Use $k = 1600$ N/m and $\delta = 12$ mm.

**B4.1** Find the elastic energy stored in the squeezed ball.

**B4.2** A 0.075 kg ball leaves the shooter at 9.0 m/s. Find its kinetic energy.

**B4.3** Compare the two. Does the ball's "spring" launch it? If not, what is compression actually for? (Hint: look back at Part A. What does friction depend on?)

# Part C: Newton's second law with friction and springs

This part builds a simple shooter model: treat the ball as a block that slides without spinning.

$$\sum F = m\,a \qquad v^2 = v_0^2 + 2\,a\,\Delta x$$

**Read more**

The Physics Classroom: Newton's Second Law · College Physics 4.3: Newton's Second Law of Motion

**C1. Belt over a block.** A 0.075 kg block is squeezed between a fast-moving belt above it and a frictionless floor below it. The squeeze gives a normal force of 15 N, and $\mu_k = 0.80$ between the belt and the block.

**C1.1** Draw the free-body diagram. Ignore gravity for now (see C3).

**C1.2** Find the block's acceleration while the belt slips over it.

**C1.3** The block starts at rest and the belt touches it for 0.15 m. Find the exit speed, assuming the belt is still slipping at the end.

**C1.4** Now say the belt moves at only 5.0 m/s. How far does the block travel before it reaches belt speed? What is its exit speed now?

**C2. Exit speed vs. compression (sliding-block model).** Replace the 15 N in C1 with the spring result $N = k \cdot \delta$.

**C2.1** Derive the exit speed $v$ in terms of $\mu$, $k$, $\delta$, the contact length $L$, and $m$. Assume slipping the whole way.

**C2.2** Use $\mu = 0.80$, $k = 1600$ N/m, $L = 0.15$ m, $m = 0.075$ kg. Find $v$ for $\delta = 4$, 8, and 12 mm.

**C2.3** If $\delta$ is multiplied by 4, what happens to $v$? Explain using your formula.

**C3. Modeling check.** Compare the ball's weight (m = 0.075 kg) to the friction force in C1. Was it reasonable to ignore gravity? Give a rule of thumb for when it is safe to ignore a force.

# Part D: Rotation primer

A real ball spins as it goes through the shooter, so we need Newton's second law for rotation. Every rotational quantity has a linear twin.

| Linear | Rotational | Link between them |
|---|---|---|
| position x (m) | angle θ (rad) | $x = \theta \cdot r$ |
| velocity v (m/s) | angular velocity ω (rad/s) | $v = \omega \cdot r$ |
| acceleration a ($m/s^2$) | angular acceleration α ($rad/s^2$) | $a = \alpha \cdot r$ |
| mass m (kg) | moment of inertia I ($kg \cdot m^2$) | depends on shape |
| force F (N) | torque τ (N·m) | $\tau = r \cdot F$ (force at the rim, tangent to it) |
| $F = m \cdot a$ | $\tau = I \cdot \alpha$ | |
| $v = v_0 + a \cdot t$ | $\omega = \omega_0 + \alpha \cdot t$ | (constant acceleration) |

**Moment of inertia.** Moment of inertia I measures how hard it is to change an object's spin, the same way mass measures how hard it is to change its motion. It depends on both the mass and where that mass sits: the farther the mass is from the axis, the larger I is.

| Shape (spinning about its center) | Moment of inertia |
|---|---|
| Solid disk (like a flywheel) | $\frac{1}{2} \cdot m \cdot r^2$ |
| Solid sphere | $(2/5) \cdot m \cdot R^2$ |
| Thin hollow sphere | $(2/3) \cdot m \cdot R^2$ |

We write a ball's moment of inertia as $\beta \cdot m \cdot R^2$, where β is the shape number.

**Torque direction.** In this set the ball travels to the right. A torque that would turn the ball the normal rolling way (top moving forward, clockwise in our pictures) is positive; a torque that would turn it the other way is negative. To find a force's direction of turning, imagine it acting alone on the rim: which way would the ball start to spin? Two forces pointing the same way but on opposite sides of the center (top and bottom) turn the ball in opposite directions. Most textbooks, including the readings below, call counterclockwise positive. We choose forward rolling (clockwise) as positive so that v and ω are both positive for a ball moving right. Either choice works as long as you stick with it.

**Read more**

*Angular velocity, rpm, $v = \omega \cdot r$:* College Physics 6.1
*Torque:* Physics Classroom: Torque · College Physics 9.2
*$\omega = \omega_0 + \alpha \cdot t$:* College Physics 10.2
*Moment of inertia, $\tau = I \cdot \alpha$:* Physics Classroom: Rotational Inertia · Newton's 2nd Law for Rotation · College Physics 10.3

**D1. Wheel speed units.** A 96 mm diameter flywheel spins at 3000 rpm.

**D1.1** Convert to rad/s.

**D1.2** Find the speed of a point on the wheel's surface in m/s.

**D2. Spinning up a disk.** A solid disk (m = 0.30 kg, r = 0.050 m) starts at rest. A constant 0.50 N force pushes tangentially on its rim.

**D2.1** Find I, the torque, and α.

**D2.2** Find ω after 2.0 s, and convert it to rpm.

**D3. Torque on a ball.** A hollow ball has m = 0.075 kg and R = 0.0635 m, with β = 2/3.

**D3.1** Find its moment of inertia.

**D3.2** A 12 N friction force acts tangentially on its surface. Find the angular acceleration.

**D4. Torques that add or cancel.** A ball sits between a belt above and a floor below. The belt's friction pushes the top of the ball forward.

**D4.1** If the floor's friction pushes the bottom of the ball *backward*, do the two torques about the center add or cancel?

**D4.2** Do the two forces add or cancel for the ball's *linear* motion?

**D4.3** Repeat both questions if the floor's friction pushes the bottom *forward*.

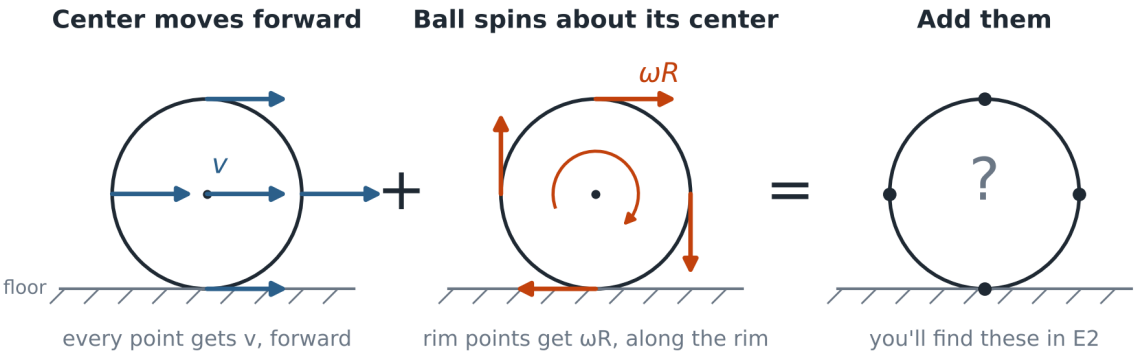
**D5. Optional (calculus).** Derive $I = \frac{1}{2} \cdot m \cdot r^2$ for a uniform disk by adding up thin rings of radius x and width dx.

# Part E: Rolling without slipping

A ball rolls without slipping when the point touching the surface is momentarily at rest relative to that surface. For a ball of radius R on still ground, that means:

$$v_{center} = \omega R \qquad a_{center} = \alpha R$$

The velocity of any point on the ball is the center's velocity plus the spin's contribution, ω·R, pointing along the rim at that point.




**Read more**
College Physics 10.4: Rotational Kinetic Energy (Example 10.10: a cylinder rolling without slipping) The car-tire example in College Physics 6.1 shows v = ω·r for a rolling wheel. Neither reading covers friction at the contact point; that's what E5 and E6 work out.

**E1. One turn of a rolling ball.** A ball with radius R = 0.0635 m rolls along the floor without slipping.

**E1.1** How far does the ball's center move when the ball rolls exactly one full turn? Give your answer in terms of R, then as a number.

**E1.2** If the ball rolls at 4.0 m/s, how long does one full turn take?

**E1.3** One full turn is 2π radians. Use your time from part 2 to find the ball's angular speed ω in rad/s.

**E1.4** Repeat part 3 with symbols: show that ω = v/R.

**E2. Speeds around a rolling ball.** A ball with R = 0.0635 m rolls on the floor at 4.0 m/s without slipping.

**E2.1** Find ω.

**E2.2** Find the velocity (size and direction) of the top point, the bottom point, and the frontmost point.

**E2.3** Sketch the ball with an arrow at each of those points.

**E3. Ball between a belt and the floor.** A ball sits between a still floor below and a belt above that moves forward at speed u. It rolls without slipping on both.

**E3.1** Let v be the center's speed and ω the spin, with ω positive when the ball turns the normal rolling way (top moving forward). Using what you found in E2, write the speed of the ball's top point and its bottom point in terms of v and ω·R.

**E3.2** “Rolling without slipping” means each contact point moves at exactly the same speed as the surface it touches. The floor is still, so set the bottom point's speed equal to 0. The belt moves at u, so set the top point's speed equal to u. Write both equations.

**E3.3** Use these two equations to solve for the velocity of the ball's center (v).

**E3.4** Find the ball's spin $\omega$ in terms of $u$ and $R$. Does it spin forward or backward relative to its direction of travel?

**E4. Connect to the shooter.** In a single-flywheel shooter, the wheel surface plays the belt and the hood plays the floor. The flywheel is 96 mm in diameter.

**E4.1** Find the fastest a ball can leave if the flywheel spins at 4000 rpm and the ball rolls without slipping on both.

**E4.2** Why can the ball never leave faster than this, no matter how hard it is squeezed? Answer in terms of slipping.

**E4.3** A two-wheel shooter grips the ball between two counter-rotating wheels at the same surface speed. Predict its maximum exit speed, and justify it with E3-style reasoning.

**E5. Spin-up.** Back to E3, but now follow the ball as it gets up to speed. The ball starts at rest. Until the top of the ball catches up, the belt slips over it, pushing the top forward with kinetic friction F. The ball rolls without slipping on the floor the whole time. Use a hollow ball, $\beta = 2/3$.

**E5.1** Draw the free-body diagram. Label the floor's friction f on the bottom of the ball. You don't know its direction yet, so assume it points backward.

**E5.2** Write Newton's second law for the ball's center.

**E5.3** Write Newton's second law for rotation, $\Sigma\tau = I\cdot\alpha$, for the ball about its center. Take forward rolling as positive.

**E5.4** Rolling on the floor means $a = R \cdot \alpha$. Use it to solve for a and f in terms of F, m, and β. (Hint: rewrite part 3 using $a = R \cdot \alpha$, then add it to part 2.)

**E5.5** Is f positive or negative? What does the sign tell you about which way the floor pushes? Explain why the floor would push that way. (Hint: imagine the floor had no friction, f = 0. Compare how fast v grows with how fast the rim speed $\omega \cdot R$ grows. Which is faster? Rolling needs $v = \omega \cdot R$, so what does the floor have to do?)

**E5.6** The belt stops slipping when the top of the ball reaches belt speed u. Using E3, show that the ball's center ends at u/2.

**E6. The slick hood.** Repeat E5, but make the floor perfectly frictionless. As before, the ball starts at rest and the belt slips over the top with kinetic friction F. Use a hollow ball, $\beta = 2/3$.

**E6.1** Draw the free-body diagram. Which forces act along the direction of motion?

**E6.2** Find the center's acceleration a and the ball's angular acceleration $\alpha$. Then multiply by R to find $R \cdot \alpha$, the rate at which the rim's spin speed $\omega \cdot R$ grows. Give a and $R \cdot \alpha$ in terms of F, m, and $\beta$.

**E6.3** The belt stops slipping when the top of the ball moves at belt speed: $v + \omega \cdot R = u$. Both $v$ and $\omega \cdot R$ start at zero and grow at constant rates, so after time $t$, $v = a \cdot t$ and $\omega \cdot R = (R \cdot \alpha) \cdot t$. Write both using your answers from part 2, substitute them into $v + \omega \cdot R = u$, and solve for the time $t$ when slipping stops. Then find $v$ at that moment as a fraction of $u$, and evaluate it for $\beta = 2/3$.

**E6.4** Compare with E5's answer, $u/2$. Did the slick floor help or hurt? Where did the “missing” speed go? Compare the ball's spin in E5 and E6.

**E6.5** Go back to your prediction in P0. Were you right? Explain in one or two sentences why friction at the floor (or hood) can *help* the ball go faster.